st231 assignment

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```
##
## Call:
## lm(formula = log(price) ~ log(x), data = diamonds)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.53162 -0.11695 -0.01271  0.12846  0.37192
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -2.46260    0.10353  -23.79   <2e-16 ***
## log(x)       5.92967    0.05687  104.27   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.168 on 605 degrees of freedom
## Multiple R-squared:  0.9473, Adjusted R-squared:  0.9472
## F-statistic: 1.087e+04 on 1 and 605 DF,  p-value: < 2.2e-16
```

a.

From summary, $M1 : \log(\text{price}) = 5.93 \cdot \log(x) - 2.46$

From Jeweller, $price \propto weight^{log1.21/log1.10}$,then $price \propto weight^2$

By geometrical background and constant density $weight \propto x^3$

Then, combine both the information from Jeweller and geometrical background, $price \propto x^6$, taking log, we get $log(price) \propto 6logx$

This theoretical coefficient of 6 is very close to the estimated coefficient 5.93 from model M1, suggesting that the fitted model is reasonable.

b.

From Jeweller, $price \propto weight^2$

By geometrical background and constant density $weight \propto x \cdot y \cdot z$

Combine these two and taking log, we get $log(price) \propto 2log(x) + 2log(y) + 2log(z)$

Then, approximate values for β_1, β_2 and β_3 are both 2.

```
##
## Call:
## lm(formula = log(price) ~ log(x) + log(y) + log(z), data = diamonds)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.4187 -0.1173 -0.0137  0.1282  0.3530
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -1.7674     0.1524 -11.595  < 2e-16 ***
## log(x)       -0.1621     0.8102  -0.200    0.842
## log(y)        4.3858     0.7800   5.623 2.87e-08 ***
## log(z)        1.7823     0.2544   7.006 6.59e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1579 on 603 degrees of freedom
## Multiple R-squared:  0.9535, Adjusted R-squared:  0.9533
## F-statistic:  4126 on 3 and 603 DF,  p-value: < 2.2e-16
```

c.

$\mathbb{E}[log(Price_j)] = \beta_0 + \beta_1 log(xj) + \beta_2 log(y_j) + \beta_3 log(z_j)$

To test H_0:$\beta_1 = 0$ vs H_1:$\beta_1 \neq 0$ we use a t-test. The number of units of observation in the dataset is $n = 607$ and the M2 model has $p = 4$ parameters. Hence, under H_0, the T-statistic, defined as $\frac{\widehat{\beta_1}}{\text{std.error}(\widehat{\beta_1})}$ has a t-distribution on $n - p = 607 - 4 = 603$ degrees of freedom. As is listed in the 2nd row of the table of coefficients, the observed value of the T-statistic is $T_{\text{obs}} = -0.200$ which has a p-value of 0.842. Thus there is insufficient evidence at the 5% level to reject the null hypothesis that $\beta_1 = 0$. We conclude that including log(x) does not have significant improvement after accounting for log(y) and log(z) in the model. Hence we do not include log(x) term in the model.

```
## Analysis of Variance Table
##
## Response: log(price)
##            Df  Sum Sq Mean Sq   F value    Pr(>F)
## log(x)      1 306.762 306.762 12296.292 < 2.2e-16 ***
```

```
## log(y)      1   0.804   0.804    32.209 2.152e-08 ***
## log(z)      1   1.224   1.224    49.081 6.587e-12 ***
## Residuals 603  15.043   0.025
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

d.

H_0: log(x) provides no additional explanatory power over the null model. H_1: log(x) contributes significant explanatory power over the null model. The test statistic is the F-statistic, which under H_0 follows an F(1,605) distribution. Observed F-value = 1.23×10^4 and corresponding p-value that is less than 0.001. We reject H_0. Hence, log(x) is highly significant when added first to the model.

This result is not contradictory to part (c). The test in (c) examines whether log(x) is significant after accounting for log(y) and log(z), while sequential ANOVA tests whether log(x) is significant alone. The high correlation between x, y, and z explains why log(x) appears insignificant in part (c) but highly significant in sequential ANOVA.

e.

From summary results of M2, estimated $\beta 1$=-0.162, $\beta 2$=4.39, $\beta 3$=1.78, while the theoretical coefficients are both 2. The estimated coefficients of model M2 are substantially different from the theoretically expected value of 2 for all three predictors. The primary reason for these counter-intuitive results is strong multicollinearity between log(x), log(y), and log(z).

In the context of round diamonds, the physical dimensions x, y and z (length, width, depth) are inherently highly correlated, as they all describe the overall size of the stone. A diamond that is large in one dimension will almost certainly be large in the others. This strong linear dependence means the three log-transformed variables carry largely overlapping information.

Numerically, this can be supported by high pairwise correlation coefficients between log(x), log(y) and log(z). Graphically, the scatterplot matrix of these variables would show strong linear associations, visually confirming the severe multicollinearity that distorts the individual slope estimates in M2.

```
##       x     y     z
## x 1.000 0.998 0.978
## y 0.998 1.000 0.976
## z 0.978 0.976 1.000
```

```
## table1: correlations between pairs of log(x),log(y) and log(z)
```

c.

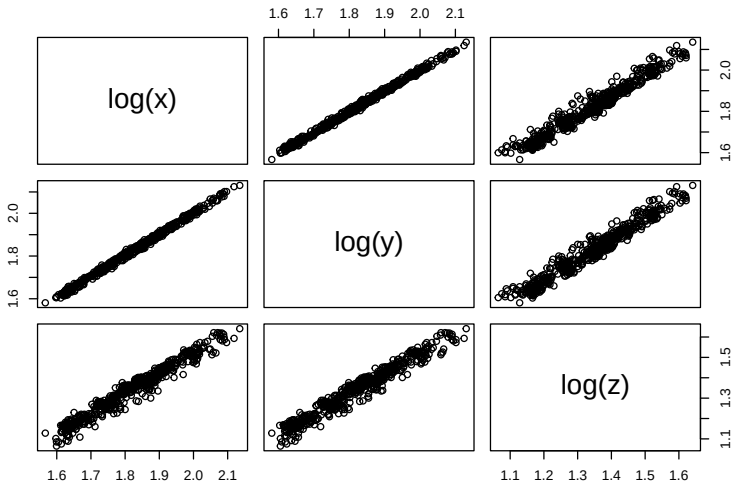


Figure 1: A scatterplot matrix of the three explanatory variables log(x), log(y) and log(z) from diamonds dataset. We see that each pair of log(x), log(y) and log(z) all have a strong linear relationship

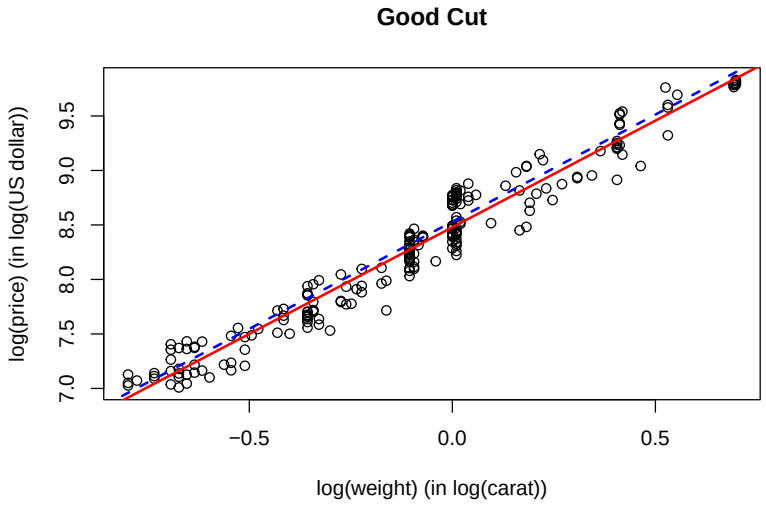


Figure 2: A scatterplot for log(weight) against log(price) for good cut dataset . The red line is the fitted regression line for for good cut dataset, and the blue line is the fitted line from model M3 using the full dataset.

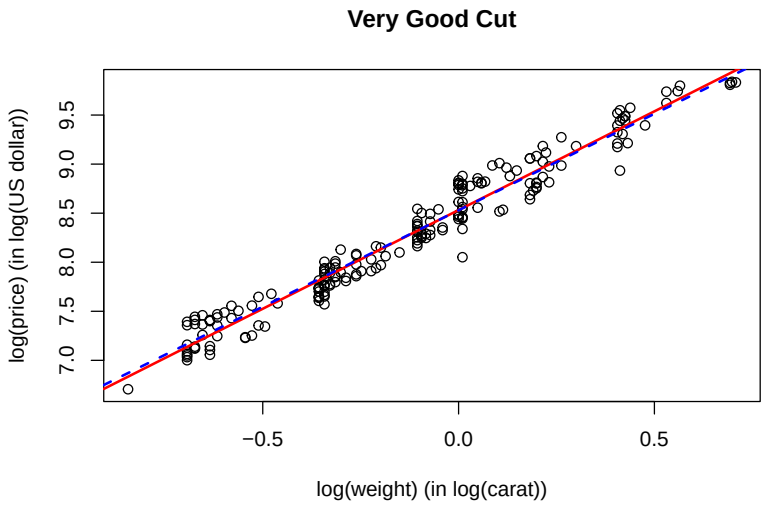


Figure 3: A scatterplot for log(weight) against log(price) for very good cut dataset. The red line is the fitted regression line for very good cut dataset, and the blue line is the fitted line from model M3 using the full dataset.

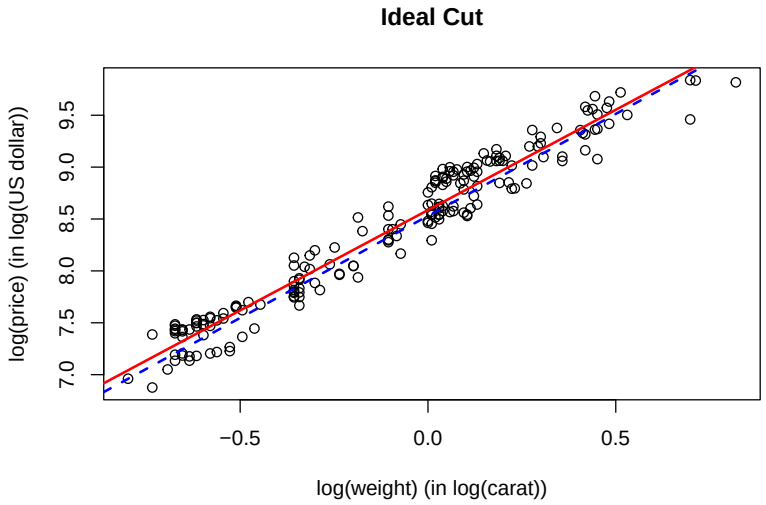


Figure 4: A scatterplot for log(weight) against log(price) for ideal cut dataset. The red line is the fitted regression line for ideal cut dataset, and the blue line is the fitted line from model M3 using the full dataset.

From the three plots for Good, Very Good and Ideal cut diamonds, we observe that: The regression lines for each cut group are similar in slope to each other and to the overall line from model M3. The intercepts differ slightly between the three cut groups. The relationship between log(price) and log(weight) is consistently linear across all levels of cut.

Merit of adding cut as a main effect: Adding cut as a main effect to model M3 is beneficial.Because the intercepts differ between cut categories, cut explains differences in the average price level at the same weight. Including cut as a main effect improves the model fit by accounting for these vertical shifts in price.

Merit of adding cut as main effect + interaction with log(weight): Adding an interaction term between cut and log(weight) is not necessary or beneficial.The slopes of the regression lines are nearly identical across the three cut groups, meaning the relationship between price and weight does not change across cut quality. The interaction term would not improve the model meaningfully.

```
## Analysis of Variance Table
##
## Model 1: log(price) ~ log(weight)
## Model 2: log(price) ~ log(weight) + cut
##   Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1    605 16.756
## 2    603 15.526  2    1.2305 23.895 1.031e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

d.

H_0: All coefficients for cut are zero. H_1: At least one coefficient for cut is non-zero. The test statistic is the F-statistic, which under H_0 follows an F(2,603) distribution. Observed F-value = 23.9 and corresponding p-value = $1.03 \times 10^{-10} < 0.05$. We reject H_0. Hence, Cut is highly significant and we include Cut variable to the model.

```
## Analysis of Variance Table
##
## Model 1: log(price) ~ log(weight) + cut
## Model 2: log(price) ~ log(weight) + cut + log(weight) * cut
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1    603 15.526
## 2    601 15.441  2  0.085072 1.6556 0.1918
```

H_0: All coefficients for the interaction cut:log(weight) are zero, given that main effects of cut and log(weight) are included. H_1: At least one interaction coefficient is non-zero , given that main effects of cut and log(weight) are included.

The test statistic is the F-statistic, which under H_0 follows an F(2,601) distribution. Observed F-value = 1.66 and corresponding p-value = 0.192 > 0.05. We do not have enough evidence to reject H_0. Hence, there is no significant evidence that the interaction improves the model, so we do not include the interaction term between cut and log(weight).

```
##
## Call:
## lm(formula = log(price) ~ cut/log(weight), data = diamonds)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.50077 -0.11321 -0.01225  0.13273  0.33827
##
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 8.47893    0.01132 749.253  < 2e-16 ***
## cutIdeal                    0.10708    0.01648   6.497 1.72e-10 ***
## cutVery Good                0.05284    0.01663   3.178  0.00156 **
## cutGood:log(weight)         1.95629    0.03054  64.062  < 2e-16 ***
## cutIdeal:log(weight)        1.93300    0.03089  62.577  < 2e-16 ***
## cutVery Good:log(weight)    2.01222    0.03214  62.615  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1603 on 601 degrees of freedom
## Multiple R-squared:  0.9523, Adjusted R-squared:  0.9519
## F-statistic:  2401 on 5 and 601 DF,  p-value: < 2.2e-16
```

e.

For Good cut: log(price)=$\beta_0 + \beta_1 log(weight)$ For Very Good cut: log(price)=$(\beta_0 + \delta_0) + (\beta_1 + \delta_1) log(weight)$

For Ideal cut: log(price)=$(\beta_0 + \gamma_0) + (\beta_1 + \gamma_1) log(weight)$

β_0 = intercept for Good cut
β_1= slope for Good cut
δ_0= intercept difference for Very Good and good cut
γ_0=intercept difference for Ideal and good cut
δ_1= slope difference for Very Good and good cut
γ_1= slope difference for Ideal and good cut

least squares estimate: β_0= 8.48,
γ_0= 0.107, δ_0= 0.0528, β_1= 1.96 ,γ_1= 1.9330-1.9563=-0.0233, δ_1= 2.0122-1.9563=0.0559

a.

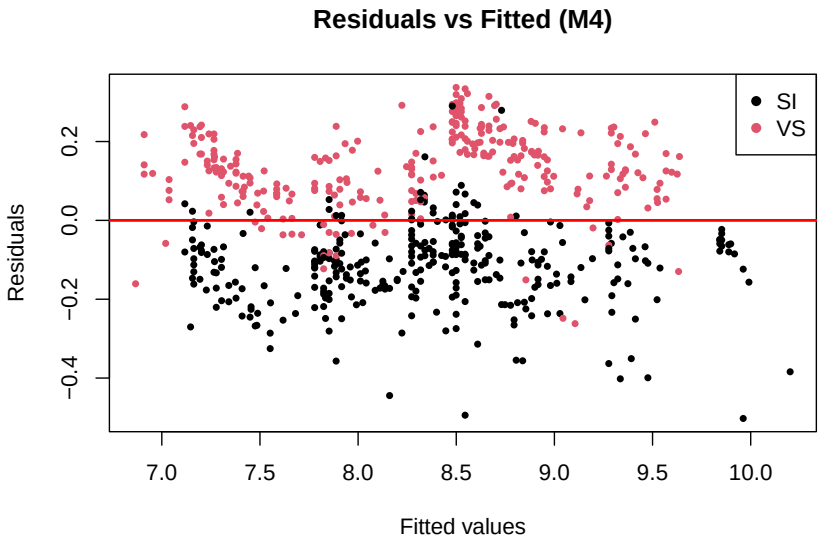


Figure 5: Residuals vs fitted values for model M4 (log(price) log(weight) + cut), points colour-coded by diamond clarity.

The residuals show a clear pattern separated by colour (clarity). Different clarity levels cluster in distinct regions of residual values. Red(VS) points cluster above the red line and black(SI) points cluster below the red line. This indicates that clarity is strongly associated with the remaining variation in price not explained by log(weight) and cut. Therefore, adding clarity as an additional predictor is likely to significantly improve the model by explaining the structured residual variation.

```
## Analysis of Variance Table
##
## Model 1: log(price) ~ log(weight) + cut
## Model 2: log(price) ~ log(weight) + cut + clarity
##   Res.Df     RSS Df Sum of Sq      F    Pr(>F)
## 1    603 15.5260
## 2    602  6.0558  1    9.4702 941.41 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

b.

Model under H0 (M4): log(price) log(weight)+cut
Model under H1 (M5): log(price) log(weight)+cut+clarity

$F = \frac{(RSS_{M4} - RSS_{M5})/(df_{M4} - df_{M5})}{RSS_{M5}/df_{M5}}$

The variable clarity has 2 levels, so it adds 2−1=1 parameters. $df_1 = 1$ Residual degrees of freedom. for M5: Total observations n=607. M5 has parameters: intercept + log(weight)

+ cut(2) + clarity(1) = 5 parameters. $df_2 = 607 - 5 = 602$. Therefore, under H_0, $F \sim F(1, 602)$
observed F = 941
p-value less than 0.001
reject H_0 There is very strong evidence that including clarity significantly improves the model for predicting log(price), after accounting for log(weight) and cut. Clarity is an important predictor of diamond prices and should be retained in the model.

c.

Yes, we could have used a two-sided t-test. From the ANOVA table, we see that the difference in degrees of freedom between M4 and M5 is 1. This means the variable clarity adds only one single parameter to the model. When comparing two nested models that differ by exactly one parameter, the partial F-test is mathematically equivalent to a two-sided t-test for that single coefficient.